JOÃO BERNARDO FERREIRA NETO

+55 16 98802-8837 | bernardojoao9@gmail.com

https://linkedin.com/in/jbnado | https://github.com/Jbnado

Professional Experience

Verzel | Full Stack Developer | Remote | April 2025 – Present

- **Context:** Full stack developer allocated to Grupo Avenida, a Brazilian fashion retail chain with 200+ stores and 6M+ private-label card customers. I work on back-office systems for the cards, credit, finance, customer service, CRM and distribution center teams. There is no designer on the team, so I own the architecture and the UI/UX of what I ship.
- **Stack:** Python, Java (Spring Boot), TypeScript (NestJS, React), Flutter, Oracle, SQL Server, Redis, RabbitMQ, BullMQ, Docker Swarm.
- **Queue and Worker Architecture:** Asynchronous processing with BullMQ, Redis and NestJS plus Python workers, a central scheduler and Docker Swarm orchestration, isolating responsibilities per job type. I built the async export module (worker, email and in-app notification) that became the project standard.
- **Bank Deposit Reconciliation:** Replaced the spreadsheet-based check the finance team did by transaction ID. Backend ingesting bank statements over FTPS, CNAB files, RabbitMQ and cron jobs (~9,300 lines), validated in production with 249 deposits and zero orphans, plus the reconciliation UI. I defined the per-bank store matching rules together with the finance team.
- **Credit Limit Engine:** Workers that apply credit limit increase and decrease campaigns of at least 100k customers, peaking at 1 million around Christmas 2025. I integrated with the card processor within its time-based rate limit windows, measured the real overnight limit and tripled throughput without hitting 429s, with per-job telemetry. It used to be manual: ad hoc queries handed off to another team.
- **Distribution Center Dashboard:** Oracle ETL, worker, API and React, fed by the WMS report through Power Automate and used daily. It replaced a spreadsheet that took 30 to 60 minutes to build. I gathered the requirements and designed the architecture with no written ticket.
- **Legacy Migration:** Migrated 9 functions from a legacy GeneXus system to the new back-office portal, back end and front end, in production since July 2026.
- **Development of Rest APIs:** Development and maintenance of **RESTful APIs** in **Python, Java (Spring Boot) and NestJS**, for high-volume transactional card systems, SAC and CRM, working with large-scale databases in **Oracle** and **SQL Server**.
- **Front-end Development:** Creation of modern and responsive interfaces with **React.js** with **Vite**, implementing reusable and optimized components for user experience.
- **Anti-fraud Blacklist:** Built with another developer: about 300k blocked IDs (fake documents, identity misuse), with fast unblocking when the real owner shows up. This data and the Overlimit feature, which grants same-day extra credit, feed the stores' point of sale directly.
- **Migrations and Database at Scale:** Consolidation and versioning of **SQL** migrations for large refactoring efforts in collaboration with DBAs, and optimization of queries and indexes on tables with millions of records to sustain performance in critical systems.
- **Observability and Reliability:** End-to-end request IDs and distributed tracing with OpenTelemetry and Jaeger across NestJS, Python services and React. Production incident diagnosis with Grafana and Loki: missing timeouts, connection pool exhaustion, retry gaps and container restarts.

- **Process and Team:** I raised the lack of agile rituals with the PO and tech lead and we introduced sprints, reviews and retros. I review the team's pull requests and onboarded a junior developer.
- **RPA Automation:** Implementation of **Robotic Process Automation** (**RPA**) processes using **Python and Playwright**, automating repetitive tasks and reducing manual processing time.

Take | Full Stack Developer | On-site | May 2023 – April 2025

- **Context:** Autonomous retail platform (smart fridges) running on a franchise model, with 5.6M orders, 444k customers and 2,680 active coolers. The back-office and app team went from 9 developers to 3 during the company's downturn; the three of us kept everything running for 6 months.
- **Stack:** Java (Spring Boot), React, Flutter, MySQL, RabbitMQ, AWS.
- **Microservices and Orders:** React front end and Java Spring Boot back end, with RabbitMQ bridging the fridges' firmware and the backend so every order was guaranteed to be processed. I improved several parts of the order processing flow.
- **Front-end Lead — Imbera and FEMSA (Coca-Cola Mexico):** I led the front end of the back office delivered to Imbera and FEMSA, built as microservices, working directly with the team in Mexico and integrating their APIs. I built the sensor dashboards, a demographic pyramid of people passing by the cooler (age estimated by AI on AWS, camera data every 5 seconds stored as Parquet) and a drag-and-drop fridge planogram with hundreds of rules.
- **Franchise Back Office:** App campaigns, banners and promotions, finance, internal and franchisee support, and user and staff management.
- **Legacy Systems Maintenance:** Refactoring and modernization of legacy backoffice in **Angular** to **React**, applying Clean Architecture and Domain-Driven Design standards, improving testability and reducing bugs.
- **Performance Optimization:** Fixed indexes on the MySQL orders table and inefficient Java logic: 60% faster load time on critical checkout pages.
- **IoT Project:** For a poultry farm client: back end and front end integrating the sensor APIs (firmware team) with the data processed by the AI team.
- **Documentation and Standardization:** Creation of design system with Storybook, API documentation with Swagger/OpenAPI and knowledge base with Wiki.js, facilitating onboarding of new developers.
- **Mentoring and Team:** Mentored 2 junior developers with code reviews and weekly feedback; both are mid-level developers today. Mediated a code style conflict between developers, agreed on a single team standard and documented it. Onboarded new developers and organized meetings.
- **AWS Infrastructure and DevOps:** Cloud infrastructure management using **AWS EC2, S3, Lambda, CloudWatch and API Gateway**, ensuring 99.8% uptime and automatic scaling based on demand with Auto Scaling Groups.
- **CI/CD and Containerization:** Implementation of CI/CD pipelines with BitBucket Pipelines and Docker, automating builds, tests and multi-environment deployments, reducing feature delivery time and ensuring zero-downtime deployments.
- **Testing Strategy**: Development of automated testing strategy with Cypress (E2E), Jest (unit) and Postman (API), achieving 80% coverage and reducing regressions in production.

Authorify | Full Stack Developer | Remote (US) | April 2022 - February 2023

- **Context:** Platform for US real estate agents: ghostwritten books, lead capture and tracking, and automated email marketing. Team of 15 developers, working fully in English.

- **Stack:** NestJS, Node.js, React, TypeScript, MongoDB, PHP, WordPress.
- **Full Stack Development:** Creation and maintenance of scalable services using **Node.js, TypeScript, NestJS, React, Jest, Cypress and MongoDB.**
- **Front-end Modernization:** Drove, together with the front-end lead, the migration from Create React App in JavaScript to TypeScript, Vite and TanStack, with hooks, Context API and clear delivery rules. 30% performance improvement.
- **MongoDB Optimization:** Reviewed schema design and indexes (every column had been indexed) and optimized the queries behind table views, cutting response time by 40%.
- **Rate-limited Integration:** Integrated with an external data source capped at 8 requests per minute to check each user's last login and trigger re-engagement emails for users inactive for 3+ months.
- **REST APIs:** Development of robust RESTful APIs with JWT authentication, data validation, and error handling.
- **Test Implementation:** Development of unit tests with Jest and E2E tests with Cypress, achieving code coverage over 85%.
- **Agile Methodology:** Active participation in Scrum ceremonies, planning, dailys, retrospectives and backlog refinement.

Target Sistemas | Full Stack Developer Trainee | Hybrid | February 2021 – April 2022

- **Front-end Foundation:** ERP software company. I joined to hand-write the HTML, CSS and JavaScript for the screens designed by UI/UX, which the back-end teams integrated into Angular.
- **Stack:** Node.js, NestJS, Express, Angular, Flutter.
- **Field Merchandiser System:** On the new projects team: point-of-sale tasks, real-time route tracking with the Google Maps API, execution and satisfaction surveys, a task kanban board and automatic PDF and Excel reports.
- **CRUD Applications:** Development of complete CRUD applications with **Express, TypeScript** and **SQL/NoSQL databases**.
- **Angular Interfaces:** Design and implementation of responsive interfaces using **Angular, RxJS and Angular Material.**
- **Adobe XD Plugin:** On my own initiative, built a plugin that generated ready-made components (buttons, tables, cards) in one click, speeding up the 2 designers' work. It was used daily until the team moved to Figma.
- **Front-end refactoring:** Refactoring of front-end code using **Sass** and **BEM methodology**, improving organization, maintainability, and visual consistency.
- **Mobile Development:** Development of cross-platform mobile application with **Flutter**.

ACADEMIC AND EDUCATIONAL PROJECT

Ribeirão Noir: The Theft of the Statue of Saint Sebastian | Solo Developer | January 2025 - August 2025

- **OOP and SOLID Architecture:** Rigorous application of **Object Oriented Programming** principles and **SOLID patterns** in the structure of the code, using design patterns (**Singleton** via Autoload, **Observer** via Signals, **Composition over Inheritance**) to ensure scalability, maintainability and low coupling.

- **Save System and State Management:** Implementation of a robust save/load system with game state serialization in JSON, global variables managed via Singleton pattern and granular tracking of player progress (clues collected, choices made, locations visited).

Technical skills

Programming Languages: TypeScript, Python, Java, JavaScript, GDScript, SQL, PL/SQL, CSS3, Sass.

Frameworks and Libraries

Backend: NestJS, Node.js (Express), Spring Boot, Flask, FastAPI.

Front-end: React.js, Next.js, Angular, Astro, Bootstrap, Material UI, Antd.

Testing: Jest, Cypress.

Database: Oracle Database, Database Design, PostgreSQL, MySQL, MongoDB, Redis, Query Optimization, Performance Tuning, Data Modeling.

DevOps and Cloud: AWS (EC2, S3, Lambda, RDS, CloudWatch), Docker, CI/CD, Git, GitHub, Azure, Nginx, Linux, Vercel, Netlify.

Automation and Integrations: Architecture, Message Queues, Microservices, RPA (Robotic Process Automation), Playwright, Redis Queue, RabbitMQ, RESTful APIs.

Tools and Methodologies: Agile/Scrum, Jira, Confluence, Figma (Dev Mode), Storybook, Wiki.js, Swagger/OpenAPI, Postman, System Design, Object-Oriented Programming (OOP), SOLID Principles, Clean Code, Design Patterns.

Languages

- **English:** Advanced (C1) – 10 months working remotely for a US company (Authorify): daily stand-ups, code reviews and technical documentation in English.
- **Portuguese:** Native

Education

- **FATEC Ribeirão Preto** | Systems Analysis & Development | 2019 - 2025